A Bialgebra Framework for Positivity in Schubert Calculus

Matthew J. Samuel

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Schubert polynomials

- Multivariate polynomials representing cohomology classes of Schubert varieties in the complete flag variety.
- Indexed by permutations, with the Schubert polynomial $\mathfrak{S}_w$ corresponding to a permutation w.
- Generalize the Schur polynomials, so their structure constants generalize the Littlewood-Richardson coefficients.
- Structure constants are nonnegative integers, but no positive combinatorial formula is known for them in general.

$$\mathfrak{S}_{41523}(x_1, x_2, x_3) = x_1^3 x_2^2 + x_1^3 x_2 x_3 + x_1^3 x_3^2$$

The "Schubert bialgebra"

Define

$$\mathbf{S} = \bigoplus_{n \geq 0} \mathbf{S}_n, \qquad \mathbf{S}_n = \mathbb{Q}[x_1, \ldots, x_n].$$

Multiplication in $\mathbf{S}$ is componentwise:

- if $f, g \in \mathbf{S}_n$, then $f \cdot g$ is the usual product in $\mathbb{Q}[x_1, \ldots, x_n]$;
- if $f \in \mathbf{S}_m$, $g \in \mathbf{S}_n$ with $m \neq n$ and $m, n > 0$, then $f \cdot g = 0$.

A monomial in n variables corresponds to a weak composition a

$$x_a = x_1^{a_1} \cdots x_n^{a_n}.$$

The coproduct is deconcatenation

$$\Delta(x_c) = \sum_{ab=c} x_a \otimes x_b$$

$$\Delta(x_1^2 x_3 x_5) = 1 \otimes x_1^2 x_3 x_5 + x_1^2 \otimes x_3 x_5 + x_1^2 x_3 \otimes x_5 + x_1^2 x_3 x_5 \otimes 1.$$

Dual algebra

We define the dual algebra $\mathcal{DS}$ to be the graded dual of **S**. For the monomial basis, which is indexed by weak compositions, we define the dual basis $[\alpha]$ for weak compositions α by

$$\langle x_\alpha, [\beta] \rangle = \delta_{\alpha,\beta}$$

We have that the dual product is concatenation:

$$[\alpha] \cdot [\beta] = [\alpha\, \beta]\,.$$

Hence $\mathcal{DS}$ is a free associative algebra (tensor algebra on the generators $[k]$, $k \geq 0$). The coproduct is

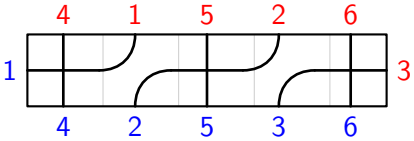
$$\Delta([\gamma]) = \sum_{\alpha+\beta=\gamma} [\alpha] \otimes [\beta]\,.$$

What do the duals of Schubs look like?

Crossing diagrams

Think of a one-row strip of $n-1$ boxes.

- Label the left edge by 1.
- Label the bottom by a permutation of the numbers $2, 3, \ldots, n$ (left to right).
- In each box, place either a *cross* ⊞ or a *bump* ◪.



Follow the strands through the row. The admissibility condition is that it cannot undo inversions, only create new ones.

A nice combinatorial basis

Let Ξ_w^k be the basis indexed by permutations w such that $w(p) < w(p+1)$ for all $p > k$ satisfying the following:

- $\Xi_w^1 = [w(1) - 1]$
- For a nonnegative integer $[p]$, we have that

$$[p] \cdot \Xi_w^k = \sum_{w'} \Xi_{w'}^{k+1}$$

  where w' is obtained from w through a valid crossing diagram with p crosses after adding 1 to each element of w and adding a new first element 1.

I can provide an explicit (but mildly cancellative) formula for Ξ_w^k in the word basis, as well as a positive formula for the product of Ξ_u^p and Ξ_v^q in terms of crossing diagrams. The coproduct structure constants are also nonnegative integers, but I do not have a positive combinatorial formula for them yet...

Taking off the mask: The dual Schubert basis

The basis satisfies

$$\langle \mathfrak{S}_u(x_1, \ldots, x_n), \Xi_v^n \rangle = \delta_{u,v}$$

where $\mathfrak{S}_u(x_1, \ldots, x_n)$ is the Schubert polynomial corresponding to the permutation u. The coefficients $c_{u,v}^w$ of the coproduct formula

$$\Delta\, \Xi_w^n = \sum_{u,v} c_{u,v}^w\, \Xi_u^n \otimes \Xi_v^n$$

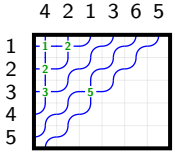
are the Littlewood-Richardson coefficients for Schubert polynomials!

Solution of the classical LR problem

- One solution of the classical LR problem (positive, combinatorial formula for multiplying Schur polynomials) is the associative algebra goverened by the Knuth relations on words of tableaux (Lascoux and Schützenberger, 1981 [2]).
- This combinatorial construction creates an algebra with clearly nonnegative structure constants
- The subalgebra generated by sums of tableaux of a given shape behave precisely like the Schur polynomials, giving the solution
- I will do this here for dual Schubert polynomials!

Bounded pipe dreams

Pipe dreams are well-studied objects introduced in the 1990s by Bergeron and Billey to give a nicer combinatorial formula for Schubert polynomials equivalent to an earlier one by Billey, Jockusch, and Stanley involving reduced words/compatible sequences.



Theorem (Bergeron, Billey, 1993 [1])

The Schubert polynomial $\mathfrak{S}_w$ is equal to

$$\sum_{R \in \mathrm{Pipe}(w)} x_{\mathrm{wt}(R)}$$

where $\mathrm{Pipe}(w)$ is the set of pipe dreams with permutation w and $\mathrm{wt}(R)$ is the weak composition whose i-th part is the number of crosses in row i of R.

Transition algorithm on bounded pipe dreams

There is an algorithm that takes a bounded pipe dream of length n with row n empty to a unique pipe dream of length $n-1$ in a way that matches the Schubert transition formula in a strong, weight-preserving, structure-preserving form.

Input: A bounded pipe dream R with $\mathrm{length}(R) = n$ and row n empty.
Output: A bounded pipe dream R' with $\mathrm{length}(R') = n-1$, $|R'| = |R|$ (same number of crossing in each row), and $w_R \overset{n}{\searrow} w_{R'}$.
if R with last row removed is a valid bounded pipe dream **then**
    return R with last row removed
else
    Locate crossings corresponding to maximal right descents of $w(R)$ until the maximal right descent is $n-1$, say at rows $i_1 < i_2 < \cdots < i_p$ with corresponding columns $j_1, j_2, \ldots, j_p$.
    $R^- \leftarrow R \setminus \{(i_1, j_1), \ldots, (i_p, j_p)\}$.
    Let $I \leftarrow (i_1, i_2, \ldots, i_p)$.
    $D \leftarrow R^- \cup \{(n,1), (n,2), \ldots, (n,p)\}$.
    $D' \leftarrow I \overset{n-1}{\rightsquigarrow} D$ (an insertion algorithm similar to existing ones for pipe dreams)
    $R' \leftarrow \{(i,j) \in D' \mid i < n\}$ as a bounded pipe dream with $n-1$ rows.
    return R'.
end if

Bounded pipe dream ring product

Given bounded pipe dreams R_1, R_2, define $R_1 \star R_2$ by taking the sum of all R obtained by

- *untransitioning* R_1 in all possible valid ways to create space while remaining a pipe dream,
- shifting R_2 downward,
- placing the shifted R_2 into the new empty region, unchanged, if doing so yields a valid pipe dream.

This allows us to lift the product of the dual algebra $\mathcal{DS}$ to a product on formal linear combinations of bounded pipe dreams.

Theorem (S, 2026+)

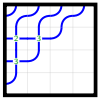
The product $\star$ is associative and gives a well-defined ring structure on the vector space Pipe *spanned by bounded pipe dreams.*

Product example

rc

$\star$

rc_2



Untransitioning of rc

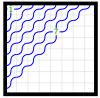
A

B

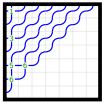
C

D

E



$rc \star rc_2 =$



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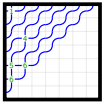

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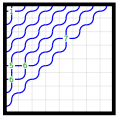

Pipe dream ring as a lift of $\mathcal{DS}$

There are two surjective linear maps from the pipe dream ring Pipe to the dual algebra $\mathcal{DS}$:

$$\pi_{\mathrm{Sch}} : \mathrm{Pipe} \twoheadrightarrow \mathcal{DS}, \qquad \pi_{\mathrm{Sch}}(G) = \Xi^{(n)}_{\mathrm{perm}(G)},$$

$$\pi_{\mathrm{wt}} : \mathrm{Pipe} \twoheadrightarrow \mathcal{DS}, \qquad \pi_{\mathrm{wt}}(G) = [\mathrm{wt}(G)]\,.$$

By virtue of the fact that $\mathcal{DS}$ is free, there is a section $\iota : \mathcal{DS} \hookrightarrow \mathrm{Pipe}$ of π_{wt} that sends $[i]$ for an integer i to the unique pipe dream r_i with $\mathrm{wt}(r_i) = (i)$. It turns out that these are all ring homomorphisms and ι is also a section of π_{Sch}.

Theorem (S, 2026+)

We have that π_{Sch} and π_{wt} are surjective ring homomorphisms to $\mathcal{DS}$ and ι is an injective ring homomorphism from $\mathcal{DS}$ to Pipe such that

$$\pi_{\mathrm{Sch}} \circ \iota = \pi_{\mathrm{wt}} \circ \iota = \mathrm{id}_{\mathcal{DS}}.$$

Littlewood-Richardson rule for (dual) Schubert polynomials

We immediately obtain a formula for the product of dual Schubert elements.

Theorem (S, 2026+)

For dual Schubert elements $\Xi_u^{n_1}$, $\Xi_v^{n_2}$, we have

$$\Xi_u^{n_1} \cdot \Xi_v^{n_2} = \sum_w d_{u,v}^w(n_1, n_2)\, \Xi_w^{n_1+n_2},$$

where, fixing pipe dreams R_u for u with $\mathrm{length}(R_u) = n_1$ and R_v for v with $\mathrm{length}(R_v) = n_2$, $d_{u,v}^w(n_1, n_2)$ is the number of bounded pipe dreams $R_w \in \mathrm{Pipe}(w)$ such that R_w has a coefficient of 1 in the product $R_u \star R_v$.
Equivalently,

$$\mathfrak{S}_w(x_1, \ldots, x_{n_1+n_2}) = \sum_{u,v} d_{u,v}^w(n_1, n_2)\, \mathfrak{S}_u(x_1, \ldots, x_{n_1})\mathfrak{S}_v(x_{n_1+1}, \ldots, x_{n_1+n_2}).$$

Forest polynomials

Any basis of the polynomial ring in n variables results in a dual basis of $\mathcal{DS}_n$. If it can be treated as a series of stable bases in n, we can analyze bases related to Schubert polynomials in this way.

- Forest polynomials (Nadeau and Tewari 2024 [3]): $\mathfrak{P}_F$ are polynomials indexed by "indexed forests" F (bijectively related to weak compositions) and can be defined by a certain equivalence relation on reduced words/compatible sequences
- It was shown that the Schubert polynomials expand positively in the forest basis. Using the equivalence relation, we can define a "forest weight" $F(R)$ for each pipe dream R, and obtain

Theorem (S, 2026+)

Let F be an indexed forest with $\mathrm{code}(F) = \mathrm{code}(w)$ for a permutation w. Then

$$\mathfrak{P}_F = \sum_{R \in \mathrm{Pipe}(w), F(R) = F} \mathrm{wt}(R)$$

New result: forest polynomial branching formula

Theorem (S, 2026+)

Write

$$\mathfrak{P}_F(x_1, \ldots, x_{n_1+n_2}) = \sum_{G,H} d_F^{G,H} \mathfrak{P}_G(x_1, \ldots, x_{n_1}) \mathfrak{P}_H(x_{n_1+1}, \ldots, x_{n_1+n_2}).$$

Then, fixing bounded pipe dreams R_G, R_H with $\mathrm{length}(R_G) = n_1$, $\mathrm{length}(R_H) = n_2$, $F(R_G) = G = \mathtt{wt}(R_G)$ and $F(R_H) = H = \mathtt{wt}(R_H)$, $d_{F,G}^H$ is the number of bounded pipe dreams R with $\mathrm{length}(R) = n_1 + n_2$ and $F(R) = F$ such that R has a coefficient of 1 in the product $R_G \star R_H$.

Corollary

Dual forest polynomials $\mathfrak{P}_F^*$ have nonnegative structure constants for the product in $\mathcal{DS}$.

Dual forest polynomial coproduct?

It is known (combinatorially, but not with a formula) that forest polynomials have nonnegative structure constants, so that dual forest polynomials $\mathfrak{P}_F^*$ have nonnegative structure constants for the coproduct in $\mathcal{DS}$.

- It is highly likely that a positive coproduct formula for $\mathfrak{P}_F^*$ in $\mathcal{DS}$ can be obtained (at least as compared to the much harder problem for Schubert polynomials) by analyzing the $\star$ product of pipe dreams with $F(R) = F$.

References I

[1] Nantel Bergeron and Sara Billey. RC-graphs and Schubert polynomials. *Experimental Math.*, 2(4):257–269, 1993.

[2] Alain Lascoux and Marcel-Paul Schützenberger. Le monoïde plaxique. In Aldo de Luca, editor, *Noncommutative structures in algebra and geometric combinatorics*. Quaderni della Ricerca Scientifica del CNR, Roma, 1981.

[3] Philippe Nadeau and Vasu Tewari. Forest polynomials and the class of the permutahedral variety. *Advances in Mathematics*, 453:109834, 2024.